79

Herbal and Over-the-Counter Medicines and the Kidney

Mark S. Segal, Xueqing Yu

The use of herbal medicines and dietary supplements continues to increase globally. An estimated one-third of adults in wealthy countries and more than 80% of the population in many low- and middle-income countries use herbal and folk medicines to promote health and manage a number of common maladies, such as colds, hay fever, indigestion, and constipation, as well as liver cirrhosis, cancer, heart disease, acquired immunodeficiency syndrome (AIDS), and diabetes.[1] With the expanding legalization of medicinal and recreational marijuana, this use is poised to increase even further. In Africa, up to 80% of the population depends on traditional medicine for primary health care; in China, herbal preparations account for up to 50% of the total consumption of pharmaceutical agents; and there has been a surge in the popularity of herbal medicine in the West. For instance, the U.S. herbal medicine market increased from \$1.6 billion in 1994 to \$9.6 billion in 2019[1,2]; complementary and alternative medicines, including Chinese herbal medicines and herbal plants, have been used by about 50% of Australians.[1]

Germany and France are the leaders in sales of over-the-counter (OTC) herbal medicines in Europe and also have large markets for prescription herbal preparations (Fig. 79.1).[3] The European Union has taken steps to regulate herbal remedies with the Traditional Herbal Medicinal Products Directive (also known as Directive 2004/24/EC), which was implemented in May of 2011. This mandates that all herbal medicinal products must obtain an authorization to market within the European Union and that herbal medicines must be manufactured under Good Manufacturing Practice.

There are at least 11,000 species of plants for medicinal use, and about 500 of them are commonly used by various ethnic groups.[1,4] These herbal plants may be used either in their primary forms or in mixtures. However, the source and composition of botanical medicines vary depending on the prevalent local practices. Herbal remedies are not tested for efficacy and safety, their ingredients are largely unknown, and there is no standardization of dosage and route of administration. Organ toxicity caused by traditional medicines is directly related to a combination of poor education, poverty, lack of medical facilities, weak or absent legislation, and widespread belief in indigenous systems of medicine in rural areas. Problems related to herbal medicines arise as a result of intrinsic toxicity, adulteration, contamination, substitution, misidentification, mistaken labeling, and unfavorable herb-drug interactions.[1,4] Increasing evidence for both adverse drug reactions and poisoning events associated with the use of herbal medicines has been reported worldwide. Herbal medicines have been found to be adulterated with synthetic drugs and other potentially toxic compounds. On the other hand, coadministration of herbal medicines with conventional drugs raises the potential for herb-drug interactions, which may cause altered drug elimination, undertreatment, and/or toxicity.[1,4,5]

HERBAL MEDICATIONS AND THE KIDNEY

The kidneys are particularly vulnerable to toxic injury because of their high blood flow rate, large endothelial surface area, high metabolic activity, active uptake by tubular cells, medullary interstitial concentration, and low urine pH. Kidney tubules are involved in active transport and urinary concentration, and therefore the local concentration of these toxins is potentially high, leading to direct injury to tubular cells. Herbal medicines may be nephrotoxic via one or more common pathogenic mechanisms, including alterations in intraglomerular hemodynamics, tubular cell toxicity, inflammation, crystal nephropathy, rhabdomyolysis, and thrombotic microangiopathy. Furthermore, potentially nephrotoxic exogenous substances such as paint thinners, turpentine, chloroxylenol, ginger, pepper, soap, vinegar, copper sulfate, and potassium permanganate are often added to herbal compounds. Drug-related nephrotoxicity is becoming more common, as more people have comorbidities that require multiple medications and more diagnostic and therapeutic procedures, all with the potential to harm kidney function. Patient-related risk factors for drug-induced nephrotoxicity include age greater than 60 years, underlying glomerular filtration rate (GFR) less than 60 mL/min/1.73 m^2, volume depletion, diabetes, heart failure, and sepsis.[6] These same risk factors also may make individuals susceptible to kidney toxicity from herbal medicines.

An overview of the potential kidney side effects of herbal medicines is shown in Table 79.1.

ARISTOLOCHIC ACID NEPHROPATHY

Aristolochic Acids

Aristolochic acids (AAs) are a family of structurally related nitrophenanthrene carboxylic acids found in herbal medicines such as *Aristolochia* spp., including *A. fangchi, A. clematitis,* and *A. manshuriensis* (Table 79.2). The predominant AAs are AAI (8-methoxy-6-nitro-phenanthro-[3,4-*d*]-1,3-dioxolo-5-carboxylic acid) and AAII (6-nitro-phenanthro-[3,4-*d*]-1,3-dioxolo-5-carboxylic acid). AAs are activated by human NAD(P)H:quinone oxidoreductase (NQO1)[7] and react with DNA to form covalent 7-(deoxyadenosin-N6-yl)aristolactam I (dA-AA) and 7-(deoxyguanosin-N2-yl)aristolactam(dG-AA) adducts. These aristolactam adducts are mutagenic.[8] After even a single dose, AAI is stably detected in the kidneys. Human aristolochic acid nephropathy (AAN) is reproducible in rodents with AA intoxication, resulting in tubular atrophy and interstitial fibrosis, leading to kidney failure.[9] The progressive tubular atrophy is related to impaired regeneration and apoptosis of proximal tubular epithelial cells, which is considered a possible mechanism of tubular epithelial cell deletion. The resident fibroblast activation plays a critical role in the process of kidney fibrosis during AA toxicity.[10] The nephrotoxic and carcinogenic

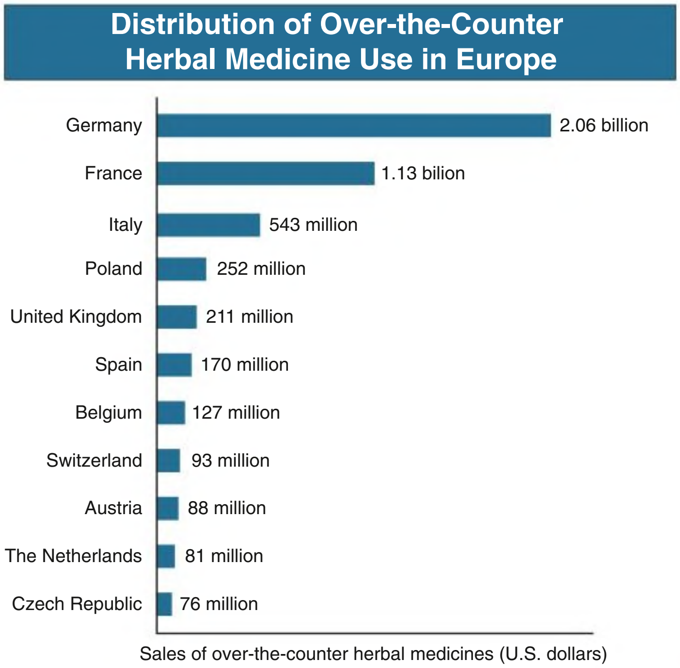

Fig. 79.1 Distribution of the $4.96 billion European market for over-the-counter herbal medicines in 2003. (Data from De Smet PA. Herbal medicine in Europe—relaxing regulatory standards. *N Engl J Med.* 2005;352:1176–1178.)

TABLE 79.1 Kidney Syndromes Induced by Herbal Medicines

Syndrome	Herbal Medicine
Hypertension	*Glycyrrhiza* spp. (Chinese herbal teas, gancao, boui-ougi-tou) *Ephedra* spp. (ma huang)
Acute tubular necrosis	Traditional African medicine: toxic plants (*Securidaca longepedunculata, Euphoria matabelensis, Callilepis laureola,* Cape aloes), or adulteration by dichromate Chinese medicine: *Taxus celebica* Morocco: *Takaout roumia* (paraphenylenediamine)
Acute interstitial nephritis	Peruvian medicine *(Uña de Gato)* Tung Shueh pills (adulterated by mefenamic acid)
Fanconi syndrome	Chinese herbs containing AAs (*Akebia* spp., boui-ougi-tou, Mokutsu) Chinese herbs adulterated by cadmium
Papillary necrosis	Chinese herbs adulterated by phenylbutazone
Chronic interstitial kidney fibrosis	Chinese herbs or Kampo containing AAs (*Aristolochia* spp., *Akebia* spp., Mu Tong, Boui, Mokutsu)
Urinary retention	*Datura* spp., *Rhododendron molle* (atropine, scopolamine)
Kidney stones	Ma huang (ephedrine) Cranberry juice (oxalate)
Urinary tract carcinoma	Chinese herbs containing AAs

AA, Aristolochic acids.
From Isnard Bagnis C, Deray G, Baumelou A, et al. Herbs and the kidney. *Am J Kidney Dis*. 2004;44:1–11.

TABLE 79.2 Botanicals Known or Suspected to Contain Aristolochic Acid and Their Common Names

Botanical Name	Common or Other Names
Aristolochia spp.	Aristolochia, Guan Mu tong, Guang Mu tong
Aristolochia acuminata (syn. *A. tagala*)	Oval leaf Dutchman's pipe
Aristolochia bracteata	Ukulwe
Aristolochia clematitis	Birthwort
Aristolochia contorta	Ma Dou Ling (fruit), Bei Ma Dou Ling (root), Tian Xian Teng (herb)
Aristolochia cymbifera	Mil homens
Aristolochia debilis (syn. *A. longa, A. recurvilabra, A. sinarum*)	Ma Dou Ling (fruit), Tian Xian Teng (herb), Qing Mu Xiang (root), Sei-Mokkou (Japanese), Birthwort, Long birthwort, Slender Dutchman's pipe
Aristolochia fangchi	Guang Fang ji (root), Fang ji, Fang chi, Makuboi (Japanese), Kou-boui (Japanese), Kwangbanggi (Korean)
Aristolochia heterophylla	Han Fang Ji
Aristolochia indica	Indian birthwort (root), Yin Du Ma Dou Ling
Aristolochia kaempferi (syn. *A. chrysops, A. feddei, A. heterophylla, A. mollis, A. setchuenensis, A. shimadai, A. thibetica, Isotrema chrysops, Isotrema heterophylla, Isotrema lasiops*)	Yellowmouth Dutchman's pipe, Zhu Sha Lian
Aristolochia macrophylla (syn. *A. sipho*)	Dutchman's pipe
Aristolochia manshuriensis (syn. *Hocquartia manshuriensis, Isotrema manshuriensis*)	Manchurian birthwort, Manchurian Dutchman's pipe (stem), Guan Mutong (stem), Kan-Mokutsu (Japanese), Mokubai (Japanese), Kwangbanggi (Korean)
Aristolochia maxima (syn. *Howardia hoffmannii*)	Maxima Dutchman's pipe, Da Ma Dou Ling
Aristolochia mollissima	Wooly Dutchman's pipe, Mian Mao Ma Dou Ling
Aristolochia moupinensis	Moupin Dutchman's pipe, Huai Tong
Aristolochia sarpentaria (syn. *A. serpentaria*)	Virginia snakeroot, Serpentaria, Virginia serpentary
Aristolochia triangularis	Triangular Dutchman's pipe, San Jiao Ma Dou Ling
Aristolochia tuberosa	Tuberous Dutchman's pipe, Kuai Jing Ma Dou Ling
Aristolochia tubiflora	Tubeflower, Dutchman's pipe, Guan Hua Ma Dou Ling
Aristolochia versicolar	Versicoloraus Dutchman's pipe, Bian Se Ma Dou Ling
Asarum canodense (syn. *A. acuminatum, A. ambiguum, A. canadense, A. furcatum, A. medium, A. parvifolium, A. reflexum, A. rubrocinctum*)	Wild ginger, Indian ginger, Canada snakeroot, false coltsfoot, colic root, heart snakeroot, Vermont snakeroot, Southern snakeroot, Jia Na Da Xi Xin
Asarum himalai(y)cum	Tanyou-saishin (Japanese)
Asarum splendens	Do-saishin (Japanese)

From Djukanović L, Marinković J, Marić I, et al. Contribution to the definition of diagnostic criteria for Balkan endemic nephropathy. *Nephrol Dial Transplant*. 2008;23(12):3932–3938.

Fig. 79.2 Traditional Chinese Pharmacy. A pharmacy selling traditional herbal remedies *(left)* including Fang Chi *(right)*. The true incidence of aristolochic acid nephropathy (AA) is largely unknown and probably underestimated because numerous ingredients known or suspected to contain AA are used in traditional medicine in India and Eastern Asia. (From Debelle FD, Vanherweghem JL, Nortier JL. Aristolochic acid nephropathy: A worldwide problem. *Kidney Int.* 2008;74:158–169.)

effects of AAs have been reported in animals; exposure to AA causes a progressive interstitial fibrosis associated with urothelial malignancy. Similar toxicities are observed in humans.

Aristolochic Acid Nephropathy

The association of kidney disease with long-term consumption of *A. fangchi* was first reported in Belgium in nine young women taking slimming preparations.[11] Since then there have been case reports and series from around the world.[12] Even though this disease may affect millions worldwide, high-quality epidemiologic data on the incidence and prevalence of AAN are lacking because of the absence of internationally agreed on diagnostic criteria and low awareness of the disease.[12] Increasing evidence supports that it is the AAs in the herbs that are responsible for their kidney toxicity (Figs. 79.2 and 79.3).[9] This syndrome was initially referred to as *Chinese herb nephropathy* and is now known by the more specific, mechanistic name of AAN.[13] The kidney toxicity of AA depends on the dosage and the duration of administration.[14] It is now clear that Balkan endemic nephropathy (BEN), a condition known for several decades without a known cause, is a form of AAN, in which there is contamination of wheat (flour) by *A. clematitis.*[9]

Definition

AAN is characterized by tubulointerstitial nephritis that progresses to fibrosis with deterioration of kidney function, ultimately leading to end-stage kidney disease (ESKD), sometimes within months after first exposure. There is a high associated risk for uroepithelial malignancy.

Epidemiology

Soon after its initial description, AAN was recognized as a global health problem.[9,11,12,14,15] Since the publication of the index cases, new cases of AAN have been reported, not only in Belgium but also worldwide.[9–16] The true incidence of AAN remains unknown and is probably underestimated because numerous ingredients known or suspected to contain AA are used in traditional medicine in China, Japan, and India (see Fig. 79.3).[9] AAN affects thousands of people living in discrete areas in Bulgaria, Bosnia, Croatia, Romania, and Serbia along the Danube River basin (Fig. 79.4), and was formerly known as BEN there.[15,17] This form of AAN typically presents in the fourth or fifth decade of life and is rarely seen in individuals younger than 20 years, presumably reflecting low-grade chronic exposure to AA.

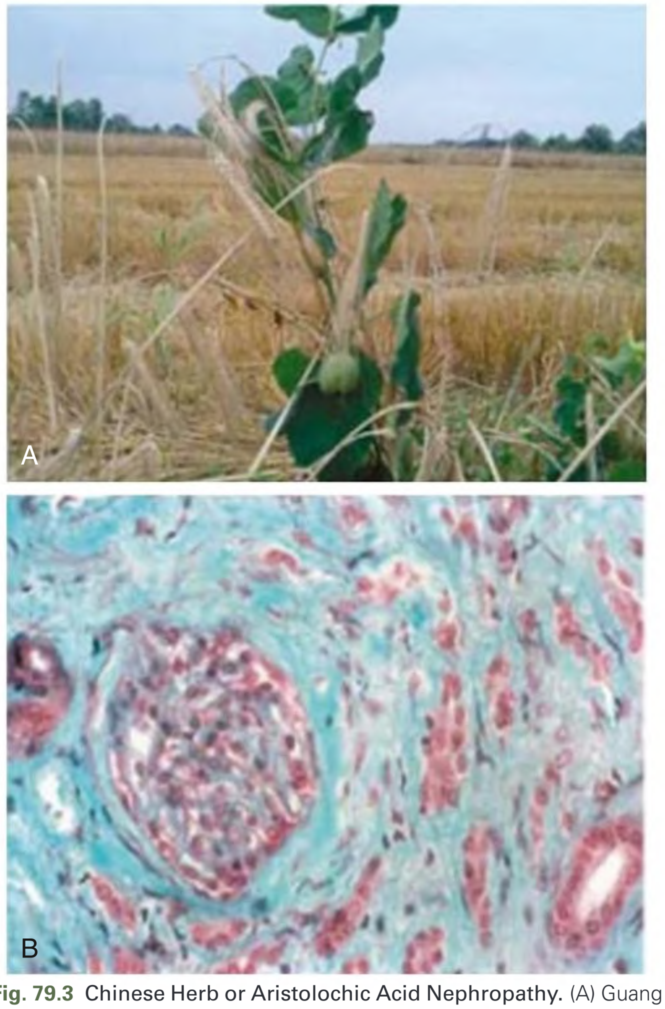

Fig. 79.3 Chinese Herb or Aristolochic Acid Nephropathy. (A) Guang Mu Tong, a Chinese herb that contains aristolochic acids. (B) Extensive paucicellular interstitial fibrosis and tubular atrophy typically found in Chinese herb or aristolochic acid nephropathy. The same pathologic appearances are seen in Balkan endemic nephropathy. (From Debelle FD, Vanherweghem JL, Nortier JL. Aristolochic acid nephropathy: a worldwide problem. *Kidney Int.* 2008;74:158–169. Courtesy Dr. Bojan Jelaković.)

Areas Where Balkan Nephropathy Is Prevalent

Fig. 79.4 **Areas Where Balkan Nephropathy is Prevalent.** The endemic areas are in *red.* (From Djukanović L, Marinković J, Marić I, et al. Contribution to the definition of diagnostic criteria for Balkan endemic nephropathy. *Nephrol Dial Transplant.* 2008;23[12]:3932–3938.)

Clinical Manifestations

The initial presentation of AAN is usually silent, with the kidney disease only discovered by routine blood testing. Occasional patients present with Fanconi syndrome or with acute kidney injury (AKI) caused by acute tubular necrosis.[18] The urinary sediment is unremarkable, and the urinalysis is initially negative for albuminuria. However, urinary excretion of low molecular weight proteins (e.g., β_2-microglobulin, cystatin C) is markedly increased, and the ratio of urinary low molecular weight protein to albumin is higher than in glomerular diseases.[19] When there is prospective monitoring in endemic areas, tubular proteinuria is usually the first manifestation. Other manifestations of tubular dysfunction include impaired acidification, decreased ammonia, increased uric acid excretion, and urine-concentrating defects with salt wasting, which may precede the decrease in GFR. The disease progresses to ESKD. Other hallmarks are that the severity of the anemia is disproportionate to the degree of kidney function impairment and that the blood pressure is usually in the normal range until the development of ESKD.

The increased incidence of uroepithelial carcinomas is characteristic of AA exposure. A case series identified upper tract urothelial carcinoma as a potent risk factor for bladder urothelial carcinoma after kidney transplantation for AAN, indicating that long-term follow-up for recurrent malignancy is potentially important (see “Treatment” section).[20]

Pathology

AAN is characterized by extensive interstitial fibrosis and tubular atrophy, which generally decreases in severity from the outer to the inner cortex (see Fig. 79.3). Early in the disease, the glomeruli are relatively spared, although at later stages there is some collapse of the capillaries and wrinkling of the basement membrane. There is endothelial cell swelling with consequent thickening of interlobular and afferent arterioles.[9,21] There are no immune deposits.

Pathogenesis

The striking association between AA exposure and uroepithelial abnormalities was first noted in nephroureterectomy specimens from individuals with AAN removed at the time of transplantation, demonstrating moderate atypia and atypical hyperplasia (see Fig. 79.3).[9,22] An early and massive interstitial inflammation characterized by activated monocytes and macrophages and cytotoxic $CD8^+/CD103^+$ T lymphocytes seen in experimental AAN[23,24] suggests that the pathophysiology includes an immunologic element. The persistence of AA-DNA adducts in the kidneys of the Belgian women cohort is consistent with their postulated role of DNA adducts in urothelial cancer. Of AAN patients, 40% to 45% develop multifocal high-grade transitional cell carcinomas, mainly in the upper urinary tract.[16] There is a strong association between DNA adduct formation, mutation pattern, and tumor development. In animal models, oral ingestion of AA is followed by extensive formation of AA-DNA adducts in the forestomach, accompanied by the development of tumors.[20] A genome-wide search in a series of cancers from Taiwan (where a significant fraction of the population is prescribed herbal medicines containing AA) demonstrated the presence of AA-DNA adducts and other DNA signatures of AA exposure in cancers of the upper urinary tract. Of additional concern was the presence of AA-DNA adducts in cancers of the liver and kidney.[25] Transitional cell cancers from patients with AAN contain AA and a characteristic pattern of tumor protein 53 (TP53) mutations, 89% occurring at A:T pairs and 78% of these being A:T to T:A transversions. These types of transversions are not seen in transitional cell carcinomas of the renal pelvis in the absence of AA exposure. Molecular epidemiologic evidence relates urothelial carcinoma in patients with AAN to dietary exposure to AA.[26]

The link between AA and BEN, which confirmed that BEN is a form of AAN, came from the insight that in endemic regions, bread is a dietary staple and is traditionally prepared from flour made from locally grown wheat. The seeds of *A. clematitis,* a plant native to the region, were found interspersed with the harvested wheat grain, and dA-aristolactam and dG-aristolactam DNA adducts were found in the kidney cortex of patients with BEN but not in those with other chronic kidney diseases (CKDs).[26,27] Aristolactam-DNA adducts have been found in 70% of people in endemic areas with dietary exposure to AA and in 94% of patients with specific A:T to T:A mutations in *TP53.* Neither aristolactam-DNA adducts nor specific mutations were detected in tissues of those from nonendemic regions. Because AAN occurs only in a fraction of those individuals exposed to AA within the endemic region, it is assumed that a genetic susceptibility puts some individuals with a dietary exposure to AA at risk for AAN and urothelial carcinoma.[9,12]

Diagnosis

Diagnostic criteria for AAN are shown in Box 79.1. An analysis of 182 patients in Serbia showed that proteinuria, urine α_1-microglobulin, and kidney size are significant predictors of AAN, whereas kidney failure and several tubular disorders (urine specific gravity, increased fractional sodium excretion, and reduced tubular phosphate reabsorption) had a nonsignificant predictive value.[15,28]

Treatment

There have been no randomized trials of treatment for AAN. However, a pilot study in 35 AAN patients with CKD demonstrated a significant reduction in the number of patients reaching ESKD after 1 year of corticosteroid therapy.[29] Corticosteroid therapy was later confirmed, in a larger cohort with historical controls, to slow the progression of kidney failure.[30] It is suggested that corticosteroids, 1 mg/kg of predniso(ol)ne for 4 weeks followed by a maintenance dose of 0.1 mg/kg tapered every 2 weeks, be initiated in those with biopsy-proven AAN (including cases discovered in endemic regions) and estimated GFR (eGFR) greater than 20 mL/min per 1.73 m^2. It is recommended to discontinue steroid

BOX 79.1 Diagnostic Criteria for Aristolochic Acid Nephropathy, Including Balkan Endemic Nephropathy

- Epidemiologic pattern
- Known exposure to aristolochic acid–containing medicines or living in endemic Balkan settlements
  - Clinical features
  - Glomerular filtration rate (GFR) decrease
  - Proteinuria usually less than 1 g/24 h
  - Bland urinary sediment
  - Markers of kidney tubular damage (glycosuria, increased urinary excretion of β_2-microglobulin or α_1-microglobulin, and *N*-acetyl-β-D-glucosaminidase)
  - Typical kidney histology showing hypocellular cortical interstitial fibrosis decreasing from the outer to the inner cortex (if kidney biopsy feasible)
- Exclusion of other known kidney disease (e.g., chronic pyelonephritis [obstructive and atrophic], adult dominant polycystic kidney disease, glomerulonephritis)

Modified from Djukanović L, Marinković J, Marić I, et al. Contribution to the definition of diagnostic criteria for Balkan endemic nephropathy. *Nephrol Dial Transplant.* 2008;23(12):3932–3938; and Luciano RL, Perazella MA. Aristolochic acid nephropathy: epidemiology, clinical presentation, and treatment. *Drug Saf.* 2015;38(1):55–64.

therapy after 6 months if the eGFR continues to decrease. Although renin-angiotensin system blockade with angiotensin-converting enzyme inhibitors or angiotensin receptor blockers is important in managing CKD, there is no evidence that this strategy improves kidney function or delays progression in AAN.[12,28]

Because patients with AAN are at increased risk for uroepithelial malignancies, all patients with AAN should undergo biannual cytologic evaluation of urine. In those patients with AAN who do not yet require kidney replacement therapy (KRT), yearly surveillance is suggested with computed tomography imaging and ureteroscopy. In addition, prophylactic bilateral nephroureterectomy is strongly recommended for all patients with ESKD caused by AAN who are going to be recipients of a kidney transplant and subsequent immunosuppression.[31]

KIDNEY INJURY CAUSED BY OTHER MEDICINALS

Aside from AAs, the use of other herbal therapy may lead to kidney injury or various toxic insults, especially in patients with kidney disease. Various kidney syndromes have been reported after the use of medicinal plants, including acute tubular necrosis, acute interstitial nephritis, Fanconi syndrome, hypertension, papillary necrosis, chronic interstitial nephritis, and nephrolithiasis.[6,32] Herbal plants reported as the cause for kidney injury include *Securidaca longepedunculata* (or violet tree), *Euphorbia matabelensis, Crotalaria laburnifolia, Uncaria tomentosa* (cat's claw), *Lepidium meyenii* (common name: maca), *Tripterygium wilfordii* (Lei Gong Teng), licorice root *(Glycyrrhiza glabra),* irumban puli *(Averrhoa bilimbi),* cape aloe, *Callilepis laureola* (Impila), mushrooms, and djenkol beans *(Pithecellobium)*. In addition, the medicinal grass carp *(Ctenopharyngodon idellus)* gallbladder has been noted to cause hepatitis and acute kidney failure.[1,32–36]

ACUTE KIDNEY INJURY

T. wilfordii Hook F (TWHF) is a Chinese herbal medicine that has been used for over 2000 years. When made into a cream, it is used externally for treatment of arthritis and inflammatory swelling. Extracts of TWHF have been found to have immunosuppressive effects, which could successfully treat rheumatoid arthritis, lupus, minimal change disease, and other autoimmune disorders. The adverse effects of TWHF include gastrointestinal (GI) upset, infertility, and leukopenia. In addition, it has been reported that AKI, profound hypotension, and shock may occur after ingestion of an extract of TWHF. In rats, daily intragastric ingestion of an effective compound extracted from TWHF for 16 days led to proximal tubular dysfunction.[32,33] Despite the known risks of the herbal medication, *Tripterygium* glycosides, the active compound extracted from TWHF, is being investigated for treatment of diabetic nephropathy.

Cat's claw, or *Uña de Gato,* is a Peruvian herbal preparation made from *Uncaria,* a woody vine found in the Amazon basin. It has been used for treatment of cirrhosis, gastritis, gonorrhea, cancers of the female genital tract, and rheumatism. The oxindole alkaloids from the root bark of cat's claw are thought to invoke its putative anti-inflammatory effects, but other unknown substances contribute to the overall effect of cat's claw extracts. A case of reversible acute interstitial nephritis after the use of this preparation has been reported, likely an idiosyncratic allergic reaction to the remedy.[34]

Mushrooms

Most kidney diseases associated with mushrooms are a consequence of kidney failure resulting from mushroom-induced hepatic failure. However, throughout Europe and North America, there are a variety of nephrotoxic mushrooms that can be confused with edible mushrooms and can cause AKI. The *Cortinarius* spp. (*C. callisteus, C. cinnamomeus* group, *C. gentilis, C. orellanus, C. rainierensis, C. speciosissimus, C. splendens,* and *C. semisanguineus* group) are the most notorious (Fig. 79.5). The most common nephrotoxic mushroom is probably *C. gentilis.* In 2016 there were 6421 mushroom exposures reported in the United States and its territories, with two fatalities. The type of mushroom is unknown in over 80% of cases.[35] The history of mushroom ingestion may be remote, particularly with *Cortinarius* spp. Although GI symptoms are usually noted at the time of ingestion, they may not be severe enough for patients to seek medical attention, and symptoms of kidney failure may not manifest until 1 to 3 weeks after exposure. Presentation with a shorter latent period suggests a more severe toxicity and greater risk for severe kidney failure. Improvement in kidney function may occur within several weeks to months, but patients may require chronic dialysis or kidney transplantation.

A more recently described mushroom syndrome involves *Amanita smithiana* (U.S. Pacific Northwest) or *Amanita proxima* (France, Italy, Spain). It is thought that the toxin is 2-amino-4,5-hexadienoic acid.[36] Although it causes acute tubular necrosis within hours of ingestion, the clinical outcome is usually good.

Diagnosis of Acute Kidney Injury Induced by Folk Remedies

An accurate assessment of the contribution of herbs to AKI is made difficult by failure to elicit a history by the physician unfamiliar with these risks or denial by the patients because of fear of stigmatization or social pressure. It is often difficult to discount the contribution to AKI of the original illness for which the herbal medicine was prescribed. AKI may be either the sole manifestation or part of a multisystem and metabolic involvement.

Treatment

Management of AKI is usually supportive and includes volume replacement and correction of metabolic abnormalities. KRT is offered for the usual indications. About 60% of all folk remedy related AKI cases need KRT, with 25% to 75% mortality. Charcoal hemoperfusion is effective in clearing alfa-amanitin from circulation in those with poisoning from *Amanita* mushrooms.[37]

Fig. 79.5 Examples of the *Cortinarius* Species of Mushrooms. (A) *C. callisteus.* (B) *C. cinnamomeus.* (C) *C. gentilis.* (D) *C. orellanus.* (E) *C. speciosissimus.* (F) *C. splendens.* (G) *C. semisanguineus.* (All images are from Wikimedia Commons. A is attributed to Amadej Trnkoczy at Mushroom Observer. B is attributed to James Lindsey at Ecology of Commanster. C is attributed to Irene Anderson. D is attributed to Thomas Pruß. E is attributed to Rand Workman at Mushroom Observer. F is attributed to Thomas Stjernegaard Jeppesen.)

OTHER KIDNEY COMPLICATIONS OF HERBAL REMEDIES

Hypertension

Ma huang is an ephedra-containing herbal preparation used in the treatment of bronchial asthma, cold and flu symptoms, fever and chills, headaches and other aches, edema, and lack of perspiration. In Western countries, ephedrine and herbal ephedra preparations are used to promote weight loss and enhance athletic performance. Dietary supplements that contain ephedra alkaloids have been reported to induce hypertension, palpitations, tachycardia, and stroke. Prescribed ephedrine was not associated with a substantially increased risk for adverse cardiovascular outcomes in a registry-based case-crossover study.[38,39] However, ephedra may pose a serious health risk to some users, such as patients with kidney disease who are prone to hypertension.

The dried roots of the licorice plant *(G. glabra)* have been consumed for over 6000 years and are used as flavoring and sweating agents, as demulcents and expectorants in the Western world, and as antiallergic and anti-inflammatory agents in Asian countries, including China, Japan, and Korea. Licorice contains glycyrrhizin. After oral administration of licorice preparations, glycyrrhizinic acid is hydrolyzed by intestinal bacteria into glycyrrhetic acid. Glycyrrhetic acid can inhibit distal tubule 11-β-hydroxysteroid dehydrogenase-2 (11-BOHD-2), which converts the corticosteroid hormone cortisol to cortisone. Decreased activity of 11-BOHD-2 leads to an excess of cortisol and an overstimulation of the mineralocorticoid receptor, leading to sodium and water retention and increased excretion of potassium. Large doses of glycyrrhizinic acid over a prolonged period can cause pseudoaldosteronism, with hypokalemia, hypernatremia, edema, and hypertension, as well as arrhythmia and other cardiac disorders.[1] To minimize the adverse effects, it is recommended that licorice not be ingested for longer than 4 to 6 weeks.[40]

Crystalluria and Nephrocalcinosis

Many health drinks that are well tolerated by individuals with normal kidney function can cause serious problems in patients with kidney disease. One example is star fruit (carambola) juice, which can contain as much as 800 mg of oxalate in 100 mL and can provoke oxalate crystalluria. It should be noted that star fruit should also be avoided in dialysis patients and those with advanced kidney insufficiency because caramboxin, the neurotoxin found in star fruit, is excreted by the kidney and if ingested in large enough amounts can cause paresthesias, seizures, coma, and death. Sour carambola juice is a popular beverage in Taiwan. Although preparation of commercial carambola juice, by pickling and dilution, markedly reduces oxalate content, fresh juice or only mildly diluted postpickled juice may contain high quantities of oxalate. Cranberry concentrate tablets also can lead to an increase in oxaluria. Ingestion of ma huang drinks have been reported to lead to kidney stones, and the use of ephedrine and guaifenesin individually or in combination causes more than 35% of urinary stones that are related to pharmaceutical metabolites and 0.1% of all urinary stones.[41]

Hyperkalemia

Dietary potassium restriction is often recommended for people with chronic kidney disease, especially because hyperkalemia is a side effect of many medications used to slow the progression of kidney disease. Common foods high in potassium include oranges, bananas, tomatoes, avocadoes, and potatoes. However, some health drinks are also very high in potassium. Noni juice, often taken to increase energy, contains more potassium than any other fruit juice. The legume alfalfa *(Medicago sativa)* and the plants dandelion *(Taraxacum officinale)*, stinging nettle *(Urtica dioica)*, and horsetail *(Equisetum arvense)* all contain significant amounts of potassium[42] and may induce hyperkalemia in persons with CKD.

Urinary Obstruction

Djenkol beans or Jering *(Pithecellobium jiringa)* are broad, round, reddish beans that grow during monsoon season in Myanmar, Indonesia, and Malaysia and are considered a delicacy. The Jering seeds are extolled for their supposed ability to prevent diabetes and high blood pressure. In addition, the seeds have bladder spasmodic properties and are used as a remedy to eliminate stones from the bladder. Jering poisoning or djenkolism is characterized by spasmodic suprapubic and/or flank pain, urinary obstruction, and AKI.[42] Chronic djenkol bean consumption is associated with a fourfold higher risk for nonglomerular hematuria. The djenkol bean contains a large amount of djenkolic acid in the range of 0.3 to 1.3 g/100 g wet weight; 93% of the acid exists in a free state. Djenkolic acid crystals may lacerate kidney tissue and cause bleeding or obstruction. Obstruction of the kidney tubules by crystals of djenkol acid is a possible mechanism of acute tubular necrosis. Mild djenkolism requires pain control and hydration. Severe djenkolism, manifested by anuria and AKI, is usually managed with analgesia, aggressive hydration, and alkalinization of the urine with sodium bicarbonate to increase the solubility of djenkolic acid. Some cases of severe djenkolism with anuria do not respond to conservative therapy and require surgical intervention.[43]

Kidney Toxicity From Contaminants in Herbal Medicines

Herbal medicines may be contaminated with excessive or banned pesticides, microbial products, heavy metals, or chemical toxins or adulterated with orthodox drugs. These contaminants are related to the source of these herbal materials, whether grown in a contaminated environment, contaminated during collection, or contaminated during storage, sometimes intentionally. The presence of orthodox drugs is often a result of the intentional adulteration of the herbal remedy by the manufacturers.[44] A report found undeclared pharmaceuticals or heavy metals in 32% of Asian medicines sold in California. These included ephedrine, chlorpheniramine, methyltestosterone, phenacetin, sildenafil, corticosteroids, and fenfluramine; 10% to 15% contained lead, mercury, or arsenic. Of more than 500 Chinese drugs, approximately 10% contained undeclared drugs or heavy metals.[4]

Several case reports link these adulterations with kidney injury. The first report was a 73-year-old Malaysian woman who presented with kidney failure and bilateral papillary necrosis after having consumed 2 tablets daily of a traditional herbal preparation, freely available from Chinese medical halls, for 10 years for osteoarthritis. She denied the consumption of any other analgesics, but the analysis of the herbal preparation showed 120 mg of phenylbutazone in each tablet. The second report concerned a 34-year-old housewife taking a mixture of Chinese herbs who presented with Fanconi syndrome and nephrogenic diabetes insipidus. She had a urinary excretion of cadmium 50 times greater than normal. In another report, a patient presented with AKI with marked albuminuria, pyuria, and hematuria after a 4-week treatment with Tung Shueh pills for arthralgias. The cause was acute interstitial nephritis, and the Tung Shueh pills were found to contain both diazepam and mefenamic acid. In Morocco, the traditional *el badia,* a powder made of the seeds of *Tamarix orientalis,* is used as hair dye; however, in times when the *T. orientalis* seeds are scarce, they are replaced with *Takaout roumia,* which contains paraphenylenediamine. Inadvertent ingestion of paraphenylenediamine is an issue in a number of African countries; for example, in Morocco it is responsible for about 10% of all cases of AKI, 50% of all cases of rhabdomyolysis, 25% of intensive care unit admissions for poisonings, and two-thirds of poisoning-related deaths.[4,6]

In Thailand and other parts of Southeast Asia and India, there are reports of contamination of herbal remedies with cadmium and/or mercury. In South Africa, about 15% of herbal remedies are contaminated with uranium. In the United States, ginseng dietary supplements have been shown to contain the pesticides quintozene and hexachlorobenzene or to exceed the standard for lead content.

Herb-Drug Interactions Resulting in Adverse Kidney-Related Effects

Herbs with potential for pharmacokinetic and/or pharmacodynamic herb-drug interactions are often taken concomitantly with therapeutic drugs, without the prescribing physician's knowledge. Pharmacokinetic herb-drug interactions are caused by altered absorption, metabolism, distribution, and excretion of drugs. A frequent underlying mechanism of altered drug concentrations by concomitant herbal medicines is the induction or inhibition of hepatic and intestinal cytochrome P-450 enzymes (CYPs). There are reports of 32 drugs interacting with herbal medicines in humans. These drugs include anticoagulants (warfarin, aspirin, and phenprocoumon), sedatives and antidepressants (midazolam, alprazolam, and amitriptyline), oral contraceptives, antiretroviral agents (indinavir, ritonavir, and saquinavir), immunosuppressants (cyclosporine and tacrolimus), anticancer drugs (imatinib and irinotecan), and digoxin. Most of them are substrates for CYPs and/or P-glycoprotein, and many have narrow therapeutic indices. Toxicity arising from drug-herb interactions may be minor, moderate, or even fatal, depending on a number of factors associated with the patients, herbs, and drugs.[1]

St. John's wort, derived from the plant *Hypericum perforatum,* has been used since ancient times for depression and anxiety. It is the most common antidepressant used in Germany. St. John's wort induces a hepatic enzyme through activation of the CYP system, leading to decreased serum levels of a wide range of prescribed drugs, with possible clinically serious consequences. For example, St. John's wort ingested for 10 days by a group of healthy volunteers reduced the bioavailability of digoxin by an average of 25%. St. John's wort ingested for 2 weeks reduced the total absorption of indinavir by 50%, which is large enough to cause treatment failure. Other reports indicated significant increases in the metabolism of warfarin, theophylline, oral contraceptives, and cyclosporine. Use of St. John's wort in transplant recipients is associated with both toxicity and underdosage of calcineurin inhibitors as a result of phytochemically triggered activity changes of isoenzyme CYP3A4 metabolism and drug transport proteins.[45,46]

Gingko biloba is one of the most popular plant extracts in Europe and is approved in Germany for the treatment of dementia. *G. biloba* is composed of several flavonoids, terpenoids (e.g., ginkgolides), and organic acids believed to act synergistically as free radical scavengers. *G. biloba* should not be administered with concomitant anticoagulation or in patients with bleeding disorders. Concomitant use of aspirin or nonsteroidal antiinflammatory drugs (NSAIDs), as well as anticoagulants such as warfarin and heparin, should be avoided. Spontaneous hyphema and spontaneous bilateral subdural hematomas have been observed in patients taking *G. biloba* and are attributed to ginkgolide B, a potent inhibitor of platelet-activating factor needed to induce arachidonate-independent platelet aggregation. Hemorrhagic complications were observed often in patients administered concomitant anti-aggregate or anticoagulant therapy. However, the exact mechanism of the interaction of *G. biloba* with aspirin, warfarin, and NSAIDs remains unclear. Experimental data from rats suggest that a *G. biloba* diet markedly increased the content of CYP and activity of glutathione-*S*-transferase and markedly induced the level of CYP2B1/2, CYP3A1, and CYP3A2 messenger RNA in the liver.[47]

Most information about the toxicity of herbal medicines is found only in case reports, and precise identities of the culprit substances, toxicologic characteristics, and pathogenic mechanisms of herbal medicines remain largely unknown. Although many herbs have been used for centuries without evidence of acute kidney damage, insidious damage caused by long-term use is a concern because many herbs have not been rigorously tested for toxicity.

OVER-THE-COUNTER MEDICINES AND THE KIDNEY

In addition to the increasing popularity of OTC health foods, nutritional supplements, and medicinal products from plants or other natural sources, there is increasing consumption in many countries of pharmaceutical products bought OTC, particularly analgesics and agents for treatment of dyspepsia.

Analgesics

OTC analgesics used for fever and minor musculoskeletal symptoms are one of the most widely used classes of drugs in the developed world. NSAIDs are also used to treat inflammatory conditions; and aspirin is used prophylactically as an anticoagulant in thrombosis-related disorders. Acetaminophen (paracetamol), aspirin, ibuprofen, and naproxen are available OTC in many countries, sometimes in combinations. Although studies on the association between the long-term use of aspirin, NSAIDs, and other analgesics and ESKD have given conflicting results, many studies suggest an association between chronic ingestion of analgesics and kidney disease.

Because analgesic use is widespread, even a small percentage of increased risk for kidney disease may have major public health implications. NSAIDs inhibit cyclooxygenase and may inhibit lipoxygenase, decreasing the production of prostaglandins and leukotrienes, which account for many NSAID-induced effects on the kidney. In patients with volume depletion, renal perfusion depends on circulating prostaglandins to vasodilate the afferent arterioles, allowing more blood flow through the glomerulus. So the contribution of prostaglandins to renal homeostasis is most critical in elderly patients and those with circulatory disturbances, such as kidney or liver dysfunction, congestive heart failure, or volume depletion.[48]

Patients who warrant chronic therapy with analgesics and NSAIDs should be monitored by regular screening with dipstick urinalysis and serum creatinine for early evidence of kidney injury. Specifically, low-risk patients should be monitored within 3 months and have studies repeated every 6 to 12 months and high-risk patients should be monitored within 1 to 3 weeks and have studies repeated every 3 to 6 months. Early detection and removal of the offending agent could halt or even reverse analgesic-induced kidney injury. There is a pressing need for multicenter prospective studies to assess the true incidence of this problem and study the effects of various analgesic agents (alone and in combination) in at-risk populations.

Analgesic Nephropathy

Analgesic nephropathy, resulting from the habitual consumption over several years of compound analgesics, is characterized by renal papillary necrosis and chronic interstitial nephritis. It is an increasingly rare condition because phenacetin (the analgesic most implicated as a causative agent) is banned in many countries. Phenacetin alone was not the cause of analgesic nephropathy (in fact, phenacetin was never marketed as a single agent); it was the use of phenacetin in combination analgesics that caused the problem. A retrospective cohort study using data from the Australia and New Zealand Dialysis and Transplant Registry showed that among 31,654 patients receiving KRT over the previous 35 years, 10.2% had analgesic nephropathy, but the incidence was less than 3.5% in 2005.[49] Analgesic nephropathy is discussed further in Chapter 65.

Proton Pump Inhibitors

Proton pump inhibitors (PPIs), such as omeprazole, are now available OTC in many countries and can be associated with granulomatous allergic interstitial nephritis (see Chapter 64). Use of PPIs has been linked epidemiologically to an increased risk for CKD in the general population.

Cannabis and Synthetic Cannabinoids

With the legalization of cannabis in Canada in 2018 and the decriminalization and legalization of recreational and/or medicinal use of cannabis in some states in the United States, cannabis is now the most commonly used psychotropic drug in the United States, second only to alcohol. In herbal cannabis, the dried flowers, fruits, leaves, and stems from the female hemp plant *Cannabis sativa,* and its subspecies, there are over 100 phytocannabinoids. The major active phytocannabinoids are cannabidiol (CBD), a nonpsychoactive phytocannabinoid, and delta 9-tetrahydrocannabinol (THC), a psychoactive phytocannaminoid.[50] Phytocannabinoids, synthetic cannabinoid analogs, and endocannabinoids, such as anandamide and 2-arachidonoylglycerol, are all ligands of the cannabinoid receptor type 1 and 2 (CB1 and CB2, respectively). The principal location of CB1 is in the central nervous system, where they have an inhibitory effect on γ-aminobutyric acid (GABA). Activation of CB1 results in the neuropsychiatric effects of synthetic cannabinoids, such as elation, irritability, and anxiety. The principal location of CB2 is in tissues related to the immune system, and these receptors are suggested to have a role in the control of emesis and pain. Both CB1 and CB2 are also expressed in human podocytes and other cells of the glomerulus. Endocannabinoids, such as anandamide, activate the CB1 receptor and increase the GFR and renal blood flow in rodents, independent of changes in blood pressure. In addition, anandamide stimulates nitric oxide production and leads to inhibition of sodium transport in the thick ascending limbs of the loop of Henle.[51]

CBD is metabolized by cytochrome P450 (CYP)3A4 and CYP2C19 and is suspected to be a potent inhibitor of these pathways. All drugs metabolized by this system (Table 79.3; including tacrolimus) would have a prolonged half-life in the presence of CBD, and there have been case reports of CBD oil affecting tacrolimus levels.

Synthetic cannabinoids, like many herbal supplements, are manufactured in unregulated laboratories and accidental or purposeful contamination may occur by the manufacturers and/or distributors. The synthetic cannabinoids are solubilized, applied to dried, plant material, and sold as "synthetic marijuana," "herbal incense," "spice," and potpourri. Some synthetic cannabinoids are more powerful than THC and a number of cases of AKI have been attributed to smoking synthetic cannabinoids.[52] Although AKI may be secondary to a contaminant or nephrotoxin present in a single batch of drug, a number of the reported AKI patients used a product that was known to contain a novel fluorinated synthetic cannabinoid (XLR-11) or its metabolites were found in clinical samples of the patients.[52]

Many of the touted therapeutic benefits of CBD are for symptoms that are common in people with advanced CKD, such as pain, nausea, anxiety, and insomnia. At this time, the evidence supporting using nonsynthetic cannabinoids for symptom management is limited to the treatment of chronic neuropathic pain, with a potential to treat uremic pruritus. However, nonsynthetic cannabinoids, particularly smoked cannabis, can pose significant health risks; thus the limited substantiated therapeutic benefit of cannabis must be weighed against these risks.[53]

TABLE 79.3 Metabolism of Medications Affected by CBD

Cytochrome P450 Isoenzyme	Drugs Metabolized
CYP2C19	Amitriptyline, carisoprodol, citalopram, cyclophosphamide, diazepam, hexobarbital, imipramine, indomethacin, clomipramine, lansoprazole, mephenytoin, mephobarbital, moclobemide, nelfinavir, nilutamide, omeprazole, pantoprazole, phenobarbital, phenytoin, primidone, progesterone, proguanil, propranolol, rabeprazole, temazepam, teniposide, valproic acid, (R)-warfarin, zidovudine
CYP3A4	Acetaminophen, codeine, ciclosporin (cyclosporin), diazepam, and erythromycin, alfentanil, alprazolam, amiodarone, amlodipine, amitryptyline, aprepitant, aripiprazole, astemizole, atorvastatin, buspirone, carbamazepine, cerivastatin, chlorpheniramine, cilostazol, cisapride, clarithromycin, codeine cyclosporine A, dapsone, dexamethasone, dextromethorphan, diazepam, digoxin, diltiazem, docetaxel, domperidone, eplerenone, ergotamine, erythromycin, estradiol, etoposide, ethynylestradiol, felodipine, fentanyl, finasteride, fluoxetine, gleevec, haloperidol, halothane, hydrocortisone, indinavir, irinotecan, ivabradine, ketoconazole, lercanidipine, levonorgestrel, lidocaine, loratadine, lovastatin, methadone, metronidazole, midazolam, mifepristone, nateglinide, nelfinavir, nifedipine, nisoldipine, nitrendipine, ondansetron, omeprazole, paracetamol, pimozide, propranolol, progesterone, quetiapine, quinidine, quinine, risperidone, ritonavir, saquinavir, salmeterol, sildenafil, simvastatin, sufentanyl, tacrolimus (FK506), taxol, tamoxifen, terfenadine, testosterone, theophylline, trazodone, triazolam, valproate, warfarin, venlafaxine, verapamil, vincristine, zaleplon, ziprasidone, zolpidem

CBD, Cannabidiol.
From Tomaszewski P, Kubiak-Tomaszewska G, Pachecka J. Cytochrome P450 polymorphism: molecular, metabolic, and pharmacogenetic aspects. II. Participation of CYP isoenzymes in the metabolism of endogenous substances and drugs. *Acta Pol Pharm*. 2008;65(3);307–318.

SELF-ASSESSMENT QUESTIONS

1. A very slender 25-year-old international fashion model presented with kidney failure, minimal hypertension, and mild albuminuria. Kidney function deteriorated, and the patient received a kidney transplant. Six months later, she developed bilateral urogenital tract tumors. What was the most likely cause of her kidney disease?
   A. Analgesic nephropathy
   B. Aristolochic acid nephropathy
   C. Anorexia nervosa
   D. Heavy metal intoxication
2. A 30-year-old recent emigrant from Romania presents with a 3-month history of fatigue, dyspnea on exertion, anorexia, and nausea. On physical examination, she is thin and appears chronically unwell. Blood pressure is 160/100 mm Hg, with otherwise normal physical examination findings. Laboratory workup is significant for hematocrit 22, serum potassium 5.7 mmol/L, serum bicarbonate 19 mmol/L, serum glucose 90 mg/dL, blood urea nitrogen 80 mg/dL, and serum creatinine 4 mg/dL. Urinalysis shows 1+ glucose, 2+ protein, 20 to 30 leukocytes per high-power field. Urine culture is negative for bacteria and acid-fast bacilli. Urine protein was 500 mg/24 h; kidney ultrasound, right kidney, 9.8 cm; left kidney, 9.6 cm; and no hydronephrosis. What is the most likely cause of her impaired kidney function?
   A. Acute kidney failure from tubular necrosis
   B. Chronic glomerulonephritis
   C. Chronic tubulointerstitial nephritis (Balkan endemic nephropathy)
   D. Occult diabetic nephropathy
3. A previously noncompliant patient with stage 4 chronic kidney disease secondary to hypertension comes into your office saying he has changed his ways. He has lost weight, is taking all of his medications, and is drinking homemade star fruit (carambola) juice for its antioxidant properties. His serum creatinine is 5.1 mg/dL (previously 2.4 mg/dL). In the figure, which picture are you most likely to see on urinalysis?
   A. Cystine crystal
   B. Urate crystal
   C. Triple phosphate
   D. Oxalate crystal
4. A 52-year-old woman 5 years after kidney transplantation is taking mycophenolate mofetil and cyclosporine with stable transplant function. She reports that she just came back from a health spa and that she feels wonderful. She said she was diagnosed with depression and was started on St. John's wort by the herbalist. Which of the following transplant-related problems are you are most concerned about?
   A. Acute rejection
   B. Crystal formation in the graft
   C. Cyclosporine toxicity
   D. Hypertension

REFERENCES

1. Zhou SF, Xue CC, Yu XQ, Wang G. Metabolic activation of herbal and dietary constituents and its clinical and toxicological implications: an update. *Curr Drug Metab*. 2007;8:526–553.
2. May T, Eckl G, Reynolds VCM. US sales of herbal supplements increase by 8.6% in 2019. *HerbalGram*. 2020;127:54–69.
3. De Smet PA. Herbal medicine in Europe—relaxing regulatory standards. *N Engl J Med*. 2005;352:1176–1178.
4. Jha V, Rathi M. Natural medicines causing acute kidney injury. *Semin Nephrol*. 2008;28:416–428.
5. Naughton CA. Drug-induced nephrotoxicity. *Am Fam Physician*. 2008;78:743–750.
6. Isnard Bagnis C, Deray G, Baumelou A, et al. Herbs and the kidney. *Am J Kidney Dis*. 2004;44:1–11.
7. Martinek V, Kubickova B, Arlt VM, et al. Comparison of activation of aristolochic acid I and II with NADPH:quinone oxidoreductase, sulphotransferases and N-acetyltranferases. *Neuroendocrinol Lett*. 2011;32(suppl 1):57–70.
8. Schmeiser HH, Janssen JW, Lyons J, et al. Aristolochic acid activates ras genes in rat tumors at deoxyadenosine residues. *Cancer Res*. 1990;50:5464–5469.
9. Debelle FD, Vanherweghem JL, Nortier JL. Aristolochic acid nephropathy: a worldwide problem. *Kidney Int*. 2008;74:158–169.
10. Pozdzik AA, Salmon IJ, Debelle FD, et al. Aristolochic acid induces proximal tubule apoptosis and epithelial to mesenchymal transformation. *Kidney Int*. 2008;73:595–607.
11. Vanherweghem JL, Depierreux M, Tielemans C, et al. Rapidly progressive interstitial renal fibrosis in young women: association with slimming regimen including Chinese herbs. *Lancet*. 1993;341:387–391.
12. Gökmen MR, Cosyns JP, Arlt VM, et al. The epidemiology, diagnosis, and management of aristolochic acid nephropathy: a narrative review. *Ann Intern Med*. 2013;158(6):469–477.
13. Gillerot G, Jadoul M, Arlt VM, et al. Aristolochic acid nephropathy in a Chinese patient: time to abandon the term "Chinese herbs nephropathy". *Am J Kidney Dis*. 2001;38:E26.
14. Lai MN, Lai JN, Chen PC, et al. Increased risks of chronic kidney disease associated with prescribed Chinese herbal products suspected to contain aristolochic acid. *Nephrology (Carlton)*. 2009;14:227–234.
15. Djukanović L, Marinković J, Marić I, et al. Contribution to the definition of diagnostic criteria for Balkan endemic nephropathy. *Nephrol Dial Transplant*. 2008;23(12):3932–3938.
16. Nortier JL, Vanherweghem JL. Renal interstitial fibrosis and urothelial carcinoma associated with the use of a Chinese herb (Aristolochia fangchi). *Toxicology*. 2002; 181-182:577–580.
17. Lincoln T. Toxicology: danger in the diet. *Nature*. 2007;448:148.
18. Kong PI, Chiu YW, Kuo MC, et al. Aristolochic acid nephropathy due to herbal drug intake manifested differently as Fanconi's syndrome and end-stage renal failure—a 7-year follow-up. *Clin Nephrol*. 2008;70:537–541.
19. Kabanda A, Jadoul M, Lauwerys R, et al. Low molecular weight proteinuria in Chinese herbs nephropathy. *Kidney Int*. 1995;48:1571–1576.
20. Lemy A, Wissing KM, Rorive S, et al. Late onset of bladder urothelial carcinoma after kidney transplantation for end-stage aristolochic acid nephropathy: a case series with 15-year follow-up. *Am J Kidney Dis*. 2008;51:471–477.
21. Depierreux M, Van Damme B, Vanden Houte K, Vanherweghem JL. Pathologic aspects of a newly described nephropathy related to the prolonged use of Chinese herbs. *Am J Kidney Dis*. 1994;24:172–180.
22. Cosyns JP, Jadoul M, Squifflet JP, et al. Chinese herbs nephropathy: a clue to Balkan endemic nephropathy? *Kidney Int*. 1994;45:1680–1688.
23. Pozdzik AA, Salmon IJ, Husson CP, et al. Patterns of interstitial inflammation during the evolution of renal injury in experimental aristolochic acid nephropathy. *Nephrol Dial Transplant*. 2008;23:2480–2491.
24. Wang Y, Meng F, Arlt VM, et al. Aristolochic acid-induced carcinogenesis examined by ACB-PCR quantification of H-Ras and K-Ras mutant fraction. *Mutagenesis*. 2011;26:619–628.
25. Rosenquist TA, Grollman AP. Mutational signature of aristolochic acid: clue to the recognition of a global disease. *DNA Repair (Amst)*. 2016;44:205–211.
26. Jelaković B, Karanović S, Vuković-Lela I, et al. Aristolactam-DNA adducts are a biomarker of environmental exposure to aristolochic acid. *Kidney Int*. 2012;81:559–567.
27. Grollman AP, Shibutani S, Moriya M, et al. Aristolochic acid and the etiology of endemic (Balkan) nephropathy. *Proc Natl Acad Sci USA*. 2007;104:12129–12134.
28. Luciano RL, Perazella MA. Aristolochic acid nephropathy: epidemiology, clinical presentation, and treatment. *Drug Saf*. 2015;38(1):55–64.
29. Vanherweghem JL, Abramowicz D, Tielemans C, Depierreux M. Effects of steroids on the progression of renal failure in chronic interstitial renal fibrosis: a pilot study in Chinese herbs nephropathy. *Am J Kidney Dis*. 1996;27:209–215.
30. Martinez MC, Nortier J, Vereerstraeten P, Vanherweghem JL. Steroid therapy in chronic interstitial renal fibrosis: the case of Chinese-herb nephropathy. *Nephrol Dial Transplant*. 2002;17:2033–2034.
31. Nortier JL, Martinez MC, Schmeiser HH, et al. Urothelial carcinoma associated with the use of a Chinese herb (Aristolochia fangchi). *N Engl J Med*. 2000;342:1686–1692.
32. Asif M. A brief study of toxic effects of some medicinal herbs on kidney. *Adv Biomed Res*. 2012;1:44.
33. Dan H, Peng RX, Ao Y, Liu YH. Segment-specific proximal tubule injury in Tripterygium glycosides intoxicated rats. *J Biochem Mol Toxicol*. 2008;22:422–428.
34. Hilepo JN, Bellucci AG, Mossey RT. Acute renal failure caused by 'cat's claw' herbal remedy in a patient with systemic lupus erythematosus. *Nephron*. 1997;77:361.
35. Brandenburg WE, Ward KJ. Mushroom poisoning epidemiology in the United States. *Mycologia*. 2018;110(4):637–641.
36. Saviuc P, Danel V. New syndromes in mushroom poisoning. *Toxicol Rev*. 2006;25:199–209.
37. Splendiani G, Zazzaro D, Di Pietrantonio P, Delfino L. Continuous renal replacement therapy and charcoal plasmaperfusion in treatment of amanita mushroom poisoning. *Artif Organs*. 2000;24:305–308.
38. Bennett S, Hoffman N, Monga M. Ephedrine- and guaifenesin-induced nephrolithiasis. *J Altern Complement Med*. 2004;10:967–969.
39. Hallas J, Bjerrum L, Støvring H, Andersen M. Use of a prescribed ephedrine/caffeine combination and the risk of serious cardiovascular events: a registry-based case-crossover study. *Am J Epidemiol*. 2008;168:966–973.
40. de Klerk GJ, Nieuwenhuis MG, Beutler JJ. Hypokalaemia and hypertension associated with use of liquorice flavoured chewing gum. *BMJ*. 1997;314:731–732.
41. Leung AY, Foster S. *Encyclopedia of Common Natural Ingredients Used in Food, Drugs, and Cosmetics*. New York: John Wiley & Sons; 1996.
42. Segasothy M, Swaminathan M, Kong NC, Bennett WM. Djenkol bean poisoning (djenkolism): an unusual cause of acute renal failure. *Am J Kidney Dis*. 1995;25:63–66.
43. Bunawan NC, Rastegar A, White KP, Wang NE. Djenkolism: case report and literature review. *Int Med Case Rep J*. 2014;7:79–84.
44. Chan K. Some aspects of toxic contaminants in herbal medicines. *Chemosphere*. 2003;52:1361–1371.
45. Yang XX, Hu ZP, Duan W, et al. Drug-herb interactions: eliminating toxicity with hard drug design. *Curr Pharm Des*. 2006;12:4649–4664.
46. Ernst E. The risk-benefit profile of commonly used herbal therapies: gingko, St john's wort, ginseng, echinacea, saw palmetto, and kava. *Ann Intern Med*. 2002;136:42–53.
47. Shinozuka K, Umegaki K, Kubota Y, et al. Feeding of Gingko biloba, extract (GBE) enhances gene expression of hepatic cytochrome P-450 and attenuates the hypotensive effect of nicardipine in rats. *Life Sci*. 2002;70:2783–2792.
48. Schlondorff D. Renal prostaglandin synthesis. Sites of production and specific actions of prostaglandins. *Am J Med*. 1986;81:1–11.
49. Chang SH, Mathew TH, McDonald SP. Analgesic nephropathy and renal replacement therapy in Australia: trends, comorbidities and outcomes. *Clin J Am Soc Nephrol*. 2008;3:768–776.

50. Grotenhermen F, Russo E. *Cannabis and Cannabinoids: Pharmacology, Toxicology, and Therapeutic Potential*. Binghamton, NY: The Haworth Press; 2002.
51. Chua JT, Argueta DA, DiPatrizio NV, et al. Endocannabinoid system and the kidneys: from renal physiology to injury and disease. *Cannabis Cannabinoid Res*. 2019;4(1):10–20.
52. Centers for Disease Control and Prevention. Acute kidney injury associated with synthetic cannabinoid use— multiple states. 2012. *Morb Mortal Wkly Rep (MMWR).* 2013;62:6.
53. Ho C, Martinusen D, Clifford Lo Canadian Journal of Kidney Health and Disease. *Rev Cannabis Chronic Kidney Dis Sympt Mangt*. 2019;6:1–14.
54. Tomaszewski P, Kubiak-Tomaszewska G, Pachecka J. Cytochrome P450 polymorphism: molecular, metabolic, and pharmacogenetic aspects. II. Participation of Cyp isoenzymes in the metabolism of endogenous substances and drugs. Acta Poloniae Pharmaceutica. *Drug Res*. 2008;65(3):307–318.